Model	VQAv2			VizWiz		
	$k = 2$	$k = 4$	$k = 8$	$k = 2$	$k = 4$	$k = 8$
Frozen	-	38.2	-	-	-	-
METALM	-	45.3	-	-	-	-
Flamingo-3B	-	53.2	55.4	-	34.4	38.4
Flamingo-9B	-	**56.3**	**58.0**	-	34.9	**39.4**
KOSMOS-1 (1.6B)	**51.4**	51.8	51.4	**31.4**	**35.3**	39.0

Table 5: Few-shot visual question answering results on VQAv2 and VizWiz.VQA accuracy scores are reported.

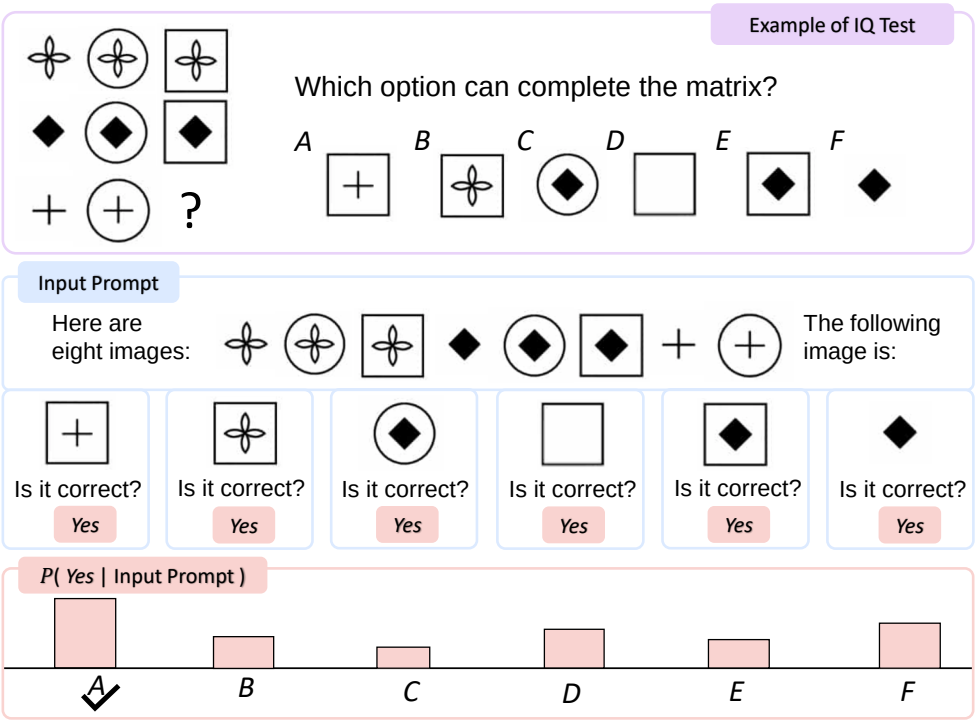

Figure 4: **Top**: An example of Raven IQ test. **Bottom**: Evaluate KOSMOS-1 on Raven IQ test. The input prompt consists of the flattened image matrix and verbal instruction. We append each candidate image to the prompt separately and query the model if it is correct. The final prediction is the candidate that motivates the model to yield the highest probability of "Yes".

4.2 IQ Test: Nonverbal Reasoning

Raven's Progressive Matrices [CJS90, JR03] is one of the most common tests to evaluate nonverbal reasoning. The capability of nonverbal reasoning is typically a reflection of an individual's intelligence quotient (IQ). Figure 4 shows an example. Given eight images presented in a 3×3 matrix, the task is to identify the following element from six similar candidates.

The models need to conduct zero-shot nonverbal reasoning without explicitly fine-tuning. The Raven IQ test is analogous to in-context learning of language models, where the difference is whether the context is nonverbal or verbal. In order to infer the answers, the models have to recognize abstract concepts and identify the underlying patterns of given images. So the IQ task is a good testbed to benchmark the nonverbal in-context learning capability.

4.2.1 Evaluation Setup

To evaluate the KOSMOS-1 on zero-shot nonverbal reasoning, we construct a dataset of the Raven IQ test. It consists of 50 examples collected from different websites[6][7][8][9]. Each example has three (i.e., 2×2 matrix), four, or eight (i.e., 3×3 matrix) given images. The goal is to predict the next one. Each instance has six candidate images with a unique correct completion. We measure accuracy scores to evaluate the models. The evaluation dataset is available at https://aka.ms/kosmos-iq50.

Figure 4 illustrates how to evaluate KOSMOS-1 on the Raven IQ test. The matrix-style images are flattened and fed into the models one-by-one. To enable the model to better understand the desired task, we also use a textual instruction *"Here are three/four/eight images:"*, *"The following image is:"*, and *"Is it correct?"* for conditioning. We append each possible candidate to the context separately and compare the probability that the model outputs "Yes" in a close-ended setting. The candidate that yields the largest probability is regarded as the prediction.

4.2.2 Results

Table 6 shows the evaluation results on the IQ test dataset. Both KOSMOS-1 with and without language-only instruction tuning achieve 5.3% and 9.3% improvement respectively over the random baseline. The results indicate that KOSMOS-1 is able to perceive abstract conceptual patterns in a nonverbal context, and then deduce the following element across multiple choices. To the best of our knowledge, it is the first time that a model can perform such zero-shot Raven IQ tests. Although there is still a large performance gap between the current model and the average level of adults, KOSMOS-1 demonstrates the potential of MLLMs to perform zero-shot nonverbal reasoning by aligning perception with language models.

Method	Accuracy
Random Choice	17%
KOSMOS-1	22%
w/o language-only instruction tuning	**26%**

Table 6: Zero-shot generalization on Raven IQ test.

4.3 OCR-Free Language Understanding

OCR-free language understanding is a task that focuses on understanding text and images without relying on Optical Character Recognition (OCR). For example, during the Rendered SST-2 task, sentences from the Stanford Sentiment Treebank [SPW+13] dataset are rendered as images. The model is asked to predict the sentiment of the text within the images. The task evaluates a model's ability to read and comprehend the meaning of words and sentences directly from the images.

4.3.1 Evaluation Setup

We evaluate OCR-free language understanding on the Rendered SST-2 [RKH+21] test set and HatefulMemes [KFM+20] validation set. We use accuracy as the metric for the Rendered SST-2 and report ROC AUC for the HatefulMemes dataset. We use the prompt *"Question: what is the sentiment of the opinion? Answer: {answer}"*, where the answer is either positive or negative for the Rendered SST-2. For the HatefulMemes task, the prompt is *"Question: does this picture contain real hate speech? Answer: {answer}"*, where the answer is either yes or no.

4.3.2 Results

As shown in Table 7, KOSMOS-1 achieves a ROC AUC of 63.9% for the HatefulMemes validation set and a test accuracy of 67.1% for the Rendered SST-2 test set. It outperforms CLIP ViT-L

[6] https://en.testometrika.com/intellectual/iq-test/
[7] https://en.testometrika.com/intellectual/iq-test-for-kids-7-to-16-year-old/
[8] https://iqpro.org/
[9] https://iqhaven.com/matrix-g

and Flamingo-9B, which achieve AUCs of 63.3% and 57.0% on the HatefulMemes task. Note that Flamingo explicitly provides OCR text into the prompt, while KOSMOS-1 does not access any external tools or resources. This indicates that KOSMOS-1 has built-in abilities to read and comprehend the text in the rendered images.

Model	HatefulMemes	Rendered SST-2
CLIP ViT-B/32	57.6	59.6
CLIP ViT-B/16	61.7	59.8
CLIP ViT-L/14	63.3	64.0
Flamingo-3B	53.7	-
Flamingo-9B	57.0	-
KOSMOS-1 (1.6B)	**63.9**	**67.1**

Table 7: Zero-shot generalization on OCR-free language understanding. We report accuracy scores.

4.4 Web Page Question Answering

Web page question answering aims at finding answers to questions from web pages. It requires the model to comprehend both the semantics and the structure of texts. The structure of the web page (such as tables, lists, and HTML layout) plays a key role in how the information is arranged and displayed. The task can help us evaluate our model's ability to understand the semantics and the structure of web pages.

4.4.1 Evaluation Setup

We compare the performance on the Web-based Structural Reading Comprehension (WebSRC) dataset [CZC+21]. For comparisons, we train a language model (LLM) on the same text corpora with the same training setup as in KOSMOS-1. The LLM takes the text extracted from the web page as input. Its template of the prompt is *"Given the context below from web page, extract the answer from the given text like this: Qusestion: Who is the publisher of this book? Answer: Penguin Books Ltd. Context: {WebText} Q: {question} A: {answer} "*, where the *{WebText}* presents the text extracted from the web page. Besides using the same prompt, KOSMOS-1 prepends the image before the prompt. Two example images from WebSRC are shown in Appendix C.3. Following the original paper [CZC+21], we use exact match (EM) and F1 scores as our evaluation metrics.

4.4.2 Results

The experimental results are summarized in Table 8. We observe that KOSMOS-1 outperforms the LLM, indicating that KOSMOS-1 can benefit from the layout and style information of web pages in images. In addition, we evaluate the performance of KOSMOS-1 without the extracted text in the prompt. It shows that extracted text has a contribution of +12.0/20.7 EM/F1 to KOSMOS-1, indicating that the benefit from modeling images does not sacrifice its language abilities.

Models	EM	F1
Using extracted text		
LLM	7.6	17.9
KOSMOS-1	**15.8**	**31.3**
Without using extracted text		
KOSMOS-1	3.8	10.6

Table 8: Zero-shot performance on WebSRC task. We report exact match (EM) and F1 scores.

4.5 Multimodal Chain-of-Thought Prompting

Chain-of-thought prompting [WWS+22] allows large language models to generate a series of reasoning steps and decompose a multi-step problem into intermediate steps, which can significantly